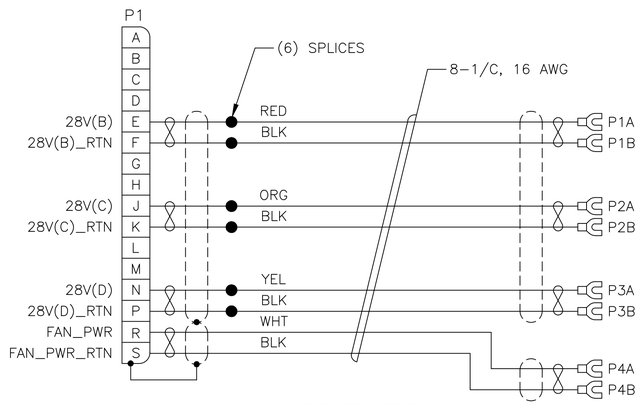
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION	MANUFACTURER
C1	1	D38999/26FG16SN	CONNECTOR, CIRCULAR PLUG, 16 POS.	QPL
C2	A/R	M39029/56-352	CONTACT, SOCKET, SIZE 16, AWG 16-20	QPL
C3	1	380HS115UC2112A3	BACKSHELL, STRAIGHT	GLENAIR
C4	8	MU14-14FLK	FORK TERMINAL, 1/4" STUD, AWG 16-14	3M
C5	A/R	M22759/11-16-0	1/C, 16 AWG, PTFE INSULATED, BLACK	QPL
C6	A/R	M22759/11-16-2	1/C, 16 AWG, PTFE INSULATED, RED	QPL
C7	A/R	M22759/11-16-3	1/C, 16 AWG, PTFE INSULATED, ORANGE	QPL
C8	A/R	M22759/11-16-4	1/C, 16 AWG, PTFE INSULATED, YELLOW	QPL
C9	A/R	M22759/11-16-9	1/C, 16 AWG, PTFE INSULATED, WHITE	QPL
C10	A/R	5011000632S	SLEEVING, EXPANDABLE, 1/4" DIA. BLACK	BENTLEY HARRIS
C11	A/R	2172	TINNED COPPER BRAID, 3/8"	ALPHA
C12	A/R	AMS-DTL-23053/13-004-0	TUBING, VITON, 1/2" DIA.	QML
C13	A/R	THFE4-365012	SHRINK TUBING, EXPANDED PREHEAT ID 0.365"	SAINT-GOBAIN
C14	A/R	2140	TINNED COPPER BRAID, 3/16"	ALPHA
C15	A/R	FMC-EXPHRZZZ-003-32	SLEEVING, EXPANDABLE, 1/8" DIA. BLACK	BENTLEY HARRIS
C16	–	–	-DELETED-	–
C17	A/R	PLT1M-M	CABLE TIE, 4" NYLON	PANDUIT
C18	A/R	TZE-211	LABEL, P-TOUCH, 1/4", BLACK ON WHITE	BROTHER
C19	6	M14BCK	TERMINAL, BUTT SPLICE, 16-14 AWG	3M
C20	A/R	P-224X.5	TAPE, KAPTON, 0.5" WIDE	PERMACEL
C21	A/R	THFE4-158010	SHRINK TUBING, EXPANDED PREHEAT ID 0.158"	SAINT-GOBAIN

LABEL LIST

ITEM	INSCRIPTION
L1	P1
L2	S1101EGA4568 SN xxxx
L3	BUS B +28VDC
L4	BUS B 28VDC RTN
L5	BUS C +28VDC
L6	BUS C 28VDC RTN
L7	BUS D +28VDC
L8	BUS D 28VDC RTN
L9	FAN PWR
L10	FAN PWR RTN



SCHEMATIC

NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4 AND FOLLOWING NOTES:
   a. TO PREVENT POTENTIAL DAMAGE (COLD FLOW) OF THE UNDERLYING WIRE INSULATION, SEPARATOR SHALL BE APPLIED OVER THE WIRE BUNDLE TO GIVE THE WIRE CONTINUOUS PROTECTION.
5. FOR TENSILE PULLS SEE TENSILE PULL STRENGTH LIST. TENSILE PULL STRENGTHS PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS.
6. INSTALL CONTACTS IN ALL UNUSED POSITIONS OF CIRCULAR CONNECTOR.
7. PERFORM CONTACT RETENTION TEST PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS. CONTACTS REQUIRING A PULL TEST FOR RETENTION SHALL BE PERFORMED AS FOLLOWS: A GENTLE PULL ON THE WIRE SHALL BE USED TO DETERMINE IF THE CONTACT IS SEATED PROPERLY.
8. TERMINATE SHIELDS INTO BACKSHELL PER MANUFACTURER'S SPECIFICATION.
9. TO SAFELY SECURE CABLE IN STRAIN RELIEF INCREASE THE CABLE DIAMETER USING HEAT SHRINK TUBING (ITEM C12). MAXIMUM OF TWO (2) LAYERS.
10. TORQUE BACKSHELL (ITEM C3) TO 80 ±5 IN-LBS.
11. TEST PER NASA-STD-8739.4 PARA 18.2.6 USING THE FOLLOWING:
    CONTINUITY: < 5 OHMS
    DWV: VOLTAGE: 1500 ±75VDC
    CURRENT: < 1 MILLIAMP
    DURATION: < 1 MINUTE
    IR: VOLTAGE: 500 ±75VDC
    RESISTANCE: < 100 MEG OHMS
    DURATION: < 1 MINUTE
12. PLACE COMPLETED ASSEMBLY IN CLEAN, STATIC-SHIELDING PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER PER SAE AS478, METHOD 35D. STORE IN A PROJECT APPROVED STORAGE AREA.

CRIMP TOOL LIST

CONTACT P/N	CRIMP TOOL	POSITIONER	I/E TOOL	REF. DES.
M39029/56-352	M22520/1-01	M22520/1-04	M81969/14-03	P1
MU14-14FLK	TR-490	N/A	N/A	P1A-P4A, P1B-P4B
M14BCK	TR-490	N/A	N/A	BUTT SPLICES

TENSILE PULL STRENGTH LIST

MANUFACTURER	CONTACT P/N	WIRE AWG	LBS
QPL	M39029/56-352	16	41.0
3M	MU14-14FLK	16	50.0
3M	M14BCK	16	50.0

BACKSHELL CLAMPING TORQUE TABLE

SCREW SIZE	TORQUE RANGE	UNITS
2-56	2 ±0.5	IN-LB
4-40	4 ±0.5	IN-LB
6-32	6 ±1.0	IN-LB
8-32	8 ±1.0	IN-LB
10-32	10 ±1.0	IN-LB
.250-20	12 ±1.0	IN-LB

REV	DESCRIPTION	APPROVED
A	REVISED PER ACME-ECO-406.	MH 9/28/15 DS 9/28/15 TG 9/29/15

REVISION HISTORY

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.

APPROVED: E. McCLOSKEY 9/29/15

PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 11/26/14	QUALITY	D. SEELEY 12/2/14
REVIEWER	J. BRUEWER 12/1/14	MATERIALS	M. WILKINSON 11/26/14
MECHANICAL	A. DRAKE 11/26/14	PROJ LEAD	B. BOROWSKI 12/2/14
ELECTRICAL	T. GOBELI 11/21/14		
STRUCTURAL	C. WERNER 11/26/14		
SAFETY	W. BORRELLI 11/26/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: S2-BT3

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE

OPTICS BENCH SIMULATOR EMI RADIATED BREAK OUT HARNESS

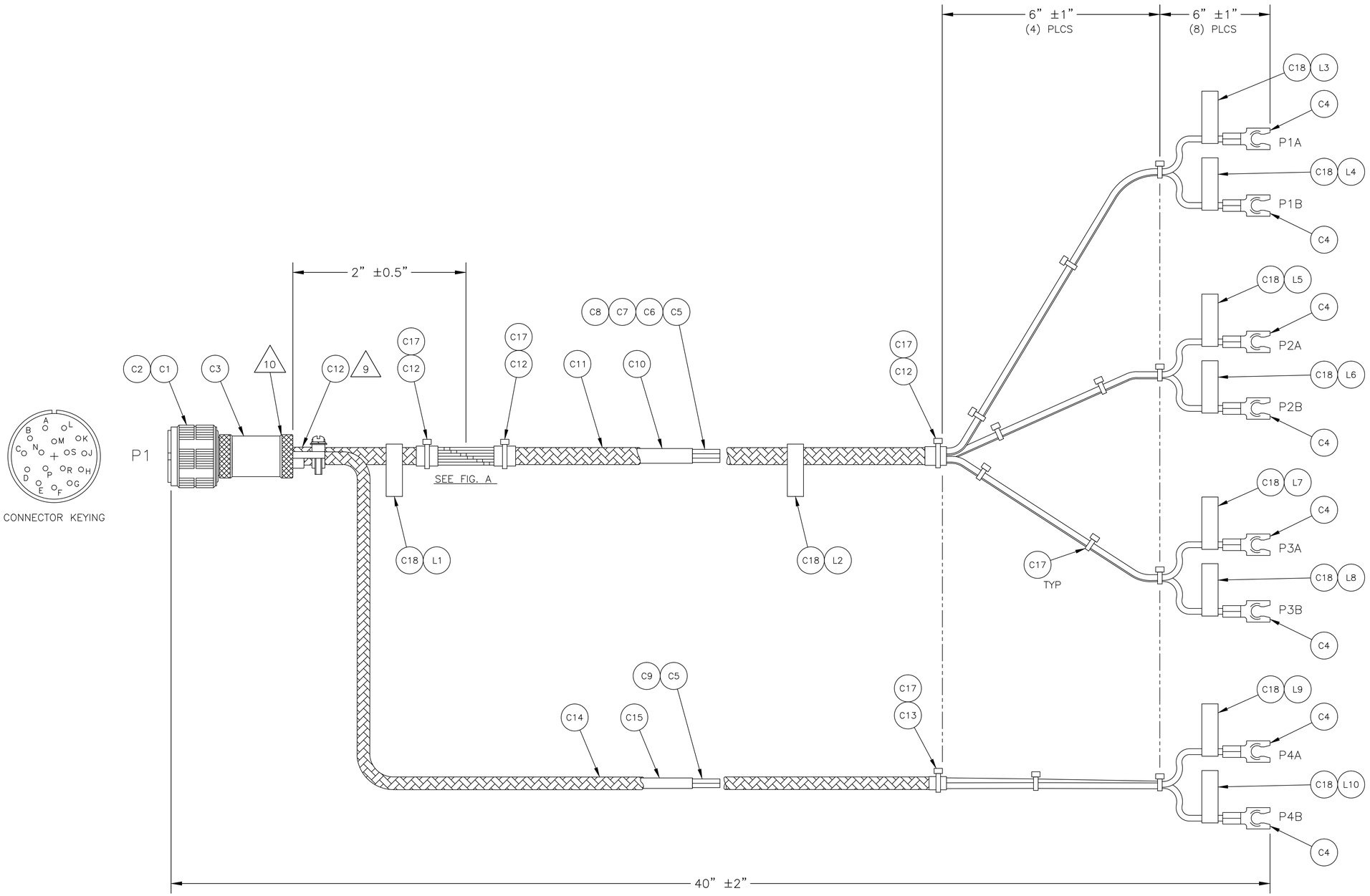
SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EGA4568	A

SCALE	NONE	INITIAL REL DATE	12/02/14	SHEET	1 OF 2

RELEASE LEVEL: RELEASED
MODEL NAME: S1101EGA4568_REV_A.DWG

AutoCAD 2012

DWG NO S1101EGA4568 SH 1 REV A



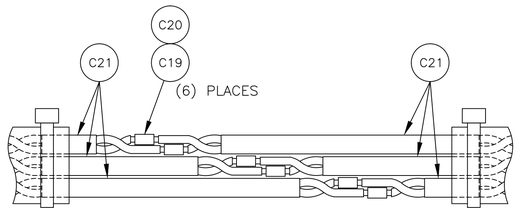


FIG. A

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EGA4568	A
SCALE NONE	INITIAL REL DATE 12/02/14	SHEET 2 OF 2	

DWG NO S1101EGA4568 SH 2 REV A

AutoCAD 2012